LESSON 6

Inference in normal populations

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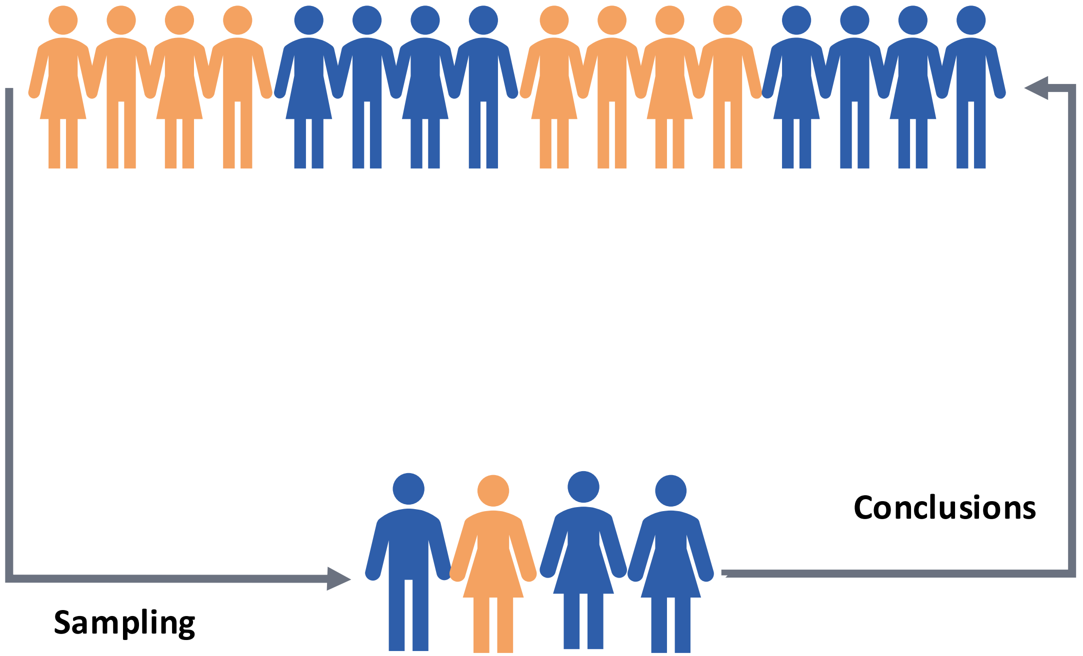
What is statistical inference

Population

Sample



Inference is extending the findings from the sample to the population.

Previous steps

Step 1: Descriptive analysis
Summarizes and visualizes sample data (position, dispersion, shape) to understand its main features before performing inferential statistics.

Step 2: Data normality
Most statistical inference techniques for continuous variables assume that the sampled populations are normally distributed.

Three complementary options:

- Use **formal statistical tests** (not in this course).
- Create a **histogram** (requires at least 40–50 data points).
- Plot on a **normal probability plot**.

Estimation

Estimating a parameter is to calculate an approximate value of it based on the values of the sample.

The estimator is a function of the values of the sample which is used to obtain approximate values of a population parameter. An estimator is, therefore, a random variable since values depend on the sample.

$\bar{x}$ **Sample mean** $\longrightarrow$ m **Population mean**

s^2 **Sample variance** $\longrightarrow$ σ^2 **Population variance**

Puntual estimation

The objective is to find out the **value of a parameter of a distribution** with greater precission and accuracy possible (average, dispersion, proportion,...)

Properties:

1) **Unbiasedness:** its expected value equals the true parameter.
2) **Consistency:** it gets closer to the true value as sample size increases.
3) **Efficiency:** it has the smallest possible variance among unbiased estimators.
4) **Sufficiency:** it uses all the relevant information in the data.

The main limitation is that a point estimate gives **no information about uncertainty**, which is why it is often complemented by confidence intervals.

Confidence intervals estimation

When we make a point estimation we also have to estimate the error, which depends on the statistics used and the sample size.

The interval estimations allow expressing in terms of probability the degree of uncertainty associated with a point estimation.

$$P\big(L_1(X) \leq \theta \leq L_2(X)\big) = 1 - \alpha$$

Where L $L_1(X)$and $L_2(X)$ are two statistics, **depending on the sample**, and $1 - \alpha$ is the confidence level. And the value (θ) between them is the true value of the parameter.

It starts by defining two competing statements: the **null hypothesis (H_0)** (a default assumption (the mean equals a given value) and the **alternative hypothesis (H_1)** (otherwise).

Then, a test statistic is computed from the data and compared against a reference distribution to assess how compatible the sample is with H_0.

$$\begin{cases} \boldsymbol{H_0}: \textit{the mean equals a given value} \\ \boldsymbol{H_1}: \textit{otherwise}. \end{cases}$$

P-value

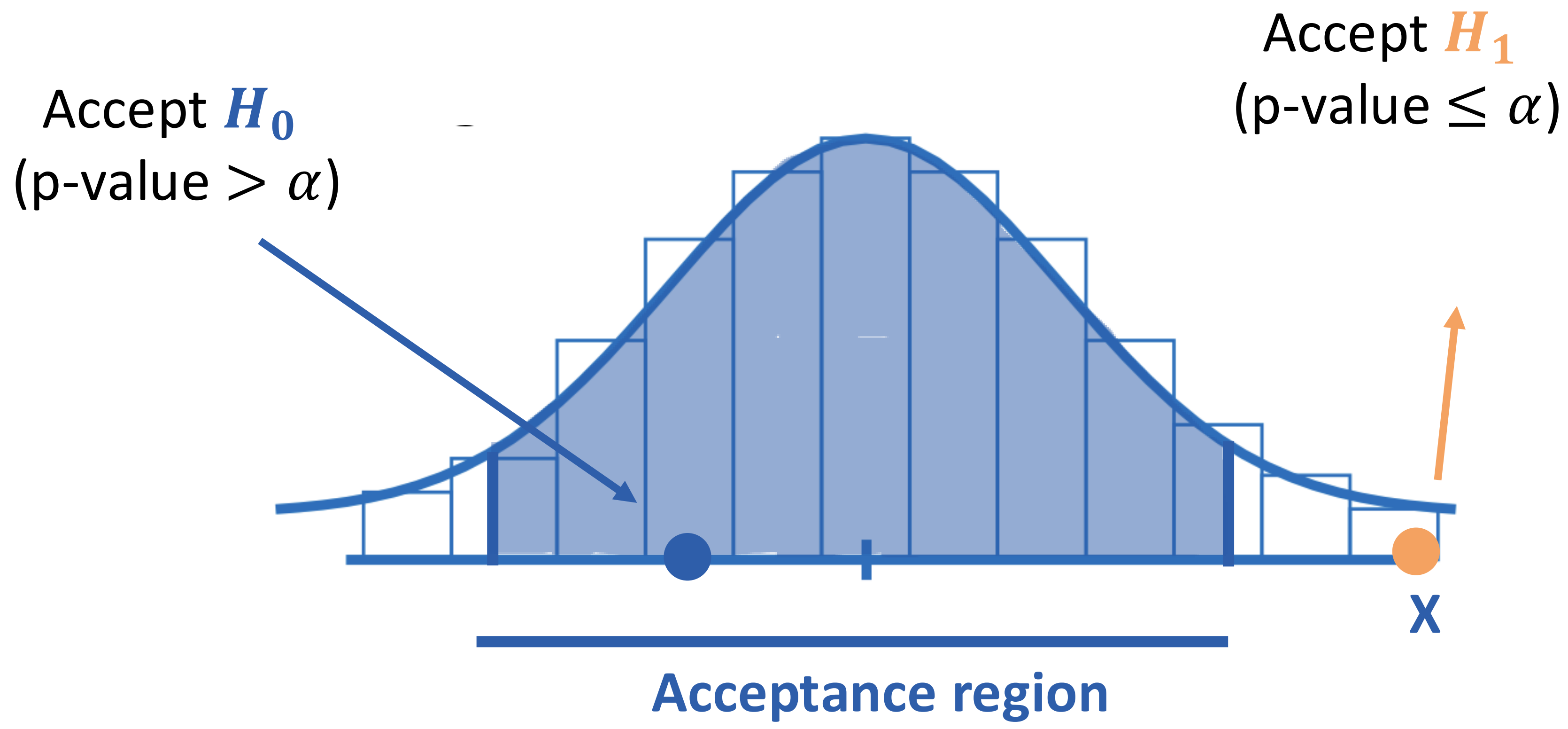
Is the probability, assuming the null hypothesis is true, of obtaining results as extreme as or more extreme than the observed data.

Accepting or rejecting the nule hypothesis

The decision rule is based on a significance level (α):

- If $p - value \ > \alpha \ \rightarrow$ accept $\boldsymbol{H_0}$ and reject $\boldsymbol{H_1}$
- If $p - value \leq \alpha \ \rightarrow$ reject $\boldsymbol{H_0}$ and accept $\boldsymbol{H_1}$

Hypothesis testing



Hypothesis testing

When conducting a statistical test on a given null hypothesis H0, two types of **erroneous conclusions** may be reached:

- **Type I** error: this occurs when $\boldsymbol{H_0}$ **is rejected** as false when it is in fact true.
- **Type II** error: this occurs when $\boldsymbol{H_0}$ **is accepted** as true when it is in fact false.

INFERENCE

REALITY		**True**	**False**
	True	Correct	**Type I error (α)**
	False	**Type II error (β)**	Correct

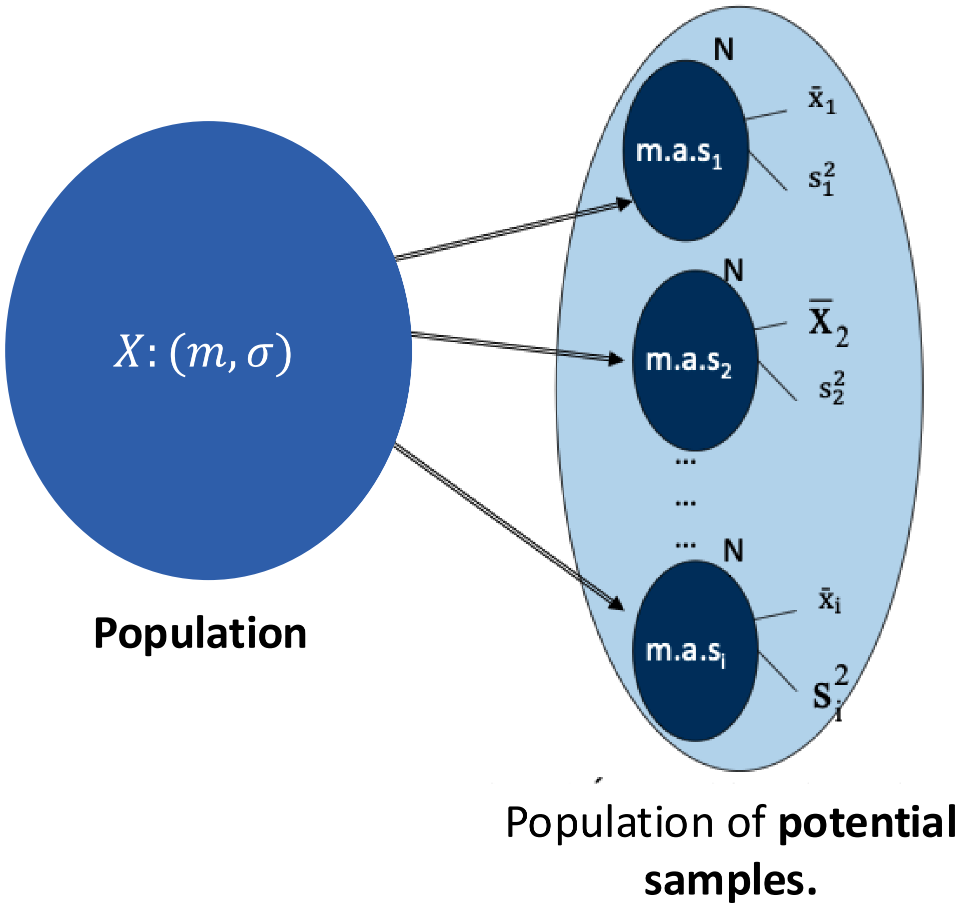
To reject an $\boldsymbol{H_0}$ hypothesis, the $\boldsymbol{\alpha}$ associated with that decision must be 'low' (0.05).

$1-\boldsymbol{\beta}$ is the power of the statistical test, the probability that this test will correctly detect that $\boldsymbol{H_0}$ is false.

Distribution of sample characteristics

It is necessary to know to what extent the mean ($\bar{x}$) and standard deviation (S) of a sample may differ from the mean (m) and standard deviation (σ) of the population from which the sample was drawn.

The sample mean and variance are **random variables** defined on the population of all possible samples.



Distribution of sample characteristics

The mean of the sample mean is the population mean

The sample mean is defined as:

$$\bar{x} = \frac{X_1+X_2+\cdots+X_N}{N} = \frac{\sum X_i}{N}$$

Each of the X_i values in the sample will be the observed value of a random variable with mean **m** and variance **σ²**.

$$E(\bar{x}) = E\left(\frac{X_1+X_2+\cdots+X_N}{N}\right) = \frac{m+m+\cdots+m}{N} = m$$

Distribution of sample characteristics

The variance of the sample mean is the population variance divided by the sample size N.

$$\sigma^2(\bar{x}) = \sigma^2\left(\frac{X_1+X_2+\cdots+X_N}{N}\right) = \frac{1}{N}\left(\sigma^2(X_1) + \sigma^2(X_2) + \cdots + \sigma^2(X_N)\right) = \frac{N\sigma^2}{N^2} = \frac{\sigma^2}{N}$$

Distribution of sample characteristics

$\bar{\boldsymbol{X}}$ is the sum of independent random variables with the same distribution

$$\bar{X} \sim N(m, \sigma^2(\bar{X}) = \frac{\sigma^2}{N})$$

When sampling from a normal population, the sample mean $\bar{x}$ is a random variable that is normally distributed with mean m (the population mean) and variance $\frac{\sigma^2}{N}$ (the population variance divided by the sample size N).

$$\frac{\bar{x} - m}{\sigma/\sqrt{N}} \sim N(0,1)$$

t-Student - Definition

If $\overline{\boldsymbol{x}}$ and $\boldsymbol{s}$ are the mean and standard deviation of a sample of size $\boldsymbol{N}$ drawn from a normal population $N(\boldsymbol{m}, \sigma)$, we can calculate the **t-Student statistic**:

$$\frac{\overline{\boldsymbol{x}} - \boldsymbol{m}}{\boldsymbol{s}/\sqrt{\boldsymbol{N}}} \sim t_{N-1}$$

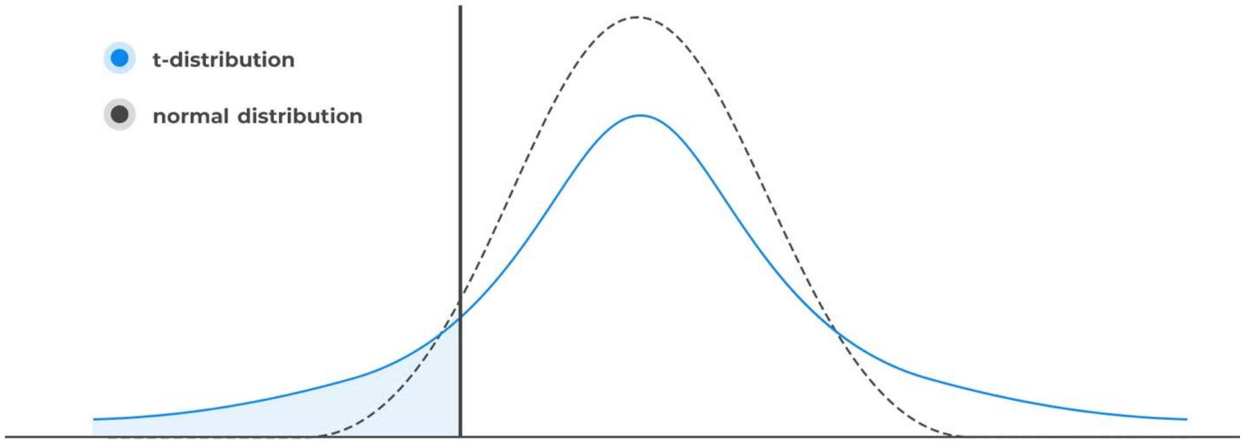
Is not the same as:

$$\frac{\overline{\boldsymbol{x}} - \boldsymbol{m}}{\boldsymbol{\sigma}/\sqrt{\boldsymbol{N}}} \sim N(0,1)$$

t-Student - Definition

Properties:

- $E(t_n) = 0$
- $\sigma^2(t_n) = \frac{n}{n-2}, when\, n > 2$
- $t_n \xrightarrow{n\to\infty} N(0,1), when\, n > 30$



t-Student – Hypothesis testing

$$\begin{cases} H_0\colon \bar{x} = m & \text{will be accepted if } \left|\frac{\bar{x}-m}{s/\sqrt{N}}\right| \leq t_{N-1}^{\alpha/2} \\ H_1\colon \bar{x} \neq m & \text{will be accepted if } \left|\frac{\bar{x}-m}{s/\sqrt{N}}\right| > t_{N-1}^{\alpha/2} \end{cases}$$

Where $t_{N-1}^{\alpha/2}$ is a value found in the **t-Student distribution table**, being $\boldsymbol{N-1}$ the degrees of freedom and $\boldsymbol{\alpha}$ the probability error.

$$P\left(\left|\frac{\bar{x}-m}{s/\sqrt{N}}\right| > t_{N-1}^{\alpha/2}\right) = \alpha$$

t-Student – Confidence Interval for m

To calculate a confidence interval with a high probability $(1-\alpha)$ of containing the unknown population mean **m** :

$$P(\bar{X} - t_{N-1}^{\alpha/2} \cdot \frac{s}{\sqrt{N}} < m < \bar{X} + t_{N-1}^{\alpha/2} \cdot \frac{s}{\sqrt{N}}) = 1-\alpha$$

$$\downarrow$$

$$IC_m = [\bar{X} - t_{N-1}^{\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{N}} \text{ , } \bar{X} + t_{N-1}^{\alpha/2} \cdot \frac{s}{\sqrt{N}}]$$

t-Student – Confidence Interval for m

$$\begin{cases} H_0:\ \bar{x} = m \\ H_1: \bar{x} \neq m \end{cases}$$

If $\mathrm{m} \in IC \rightarrow$ accept $\boldsymbol{H_0}$ and reject $\boldsymbol{H_1}$

If $m \notin IC \rightarrow$ reject $\boldsymbol{H_0}$ and accept $\boldsymbol{H_1}$

Chi-Squared distribution – Definition (χ^2)

Describes the sum of the squares of independent standard normal random variables:

$$\chi^2 = \sum_{i=1}^{n} X_i^2 \to N(0,1) \ and \ independent$$

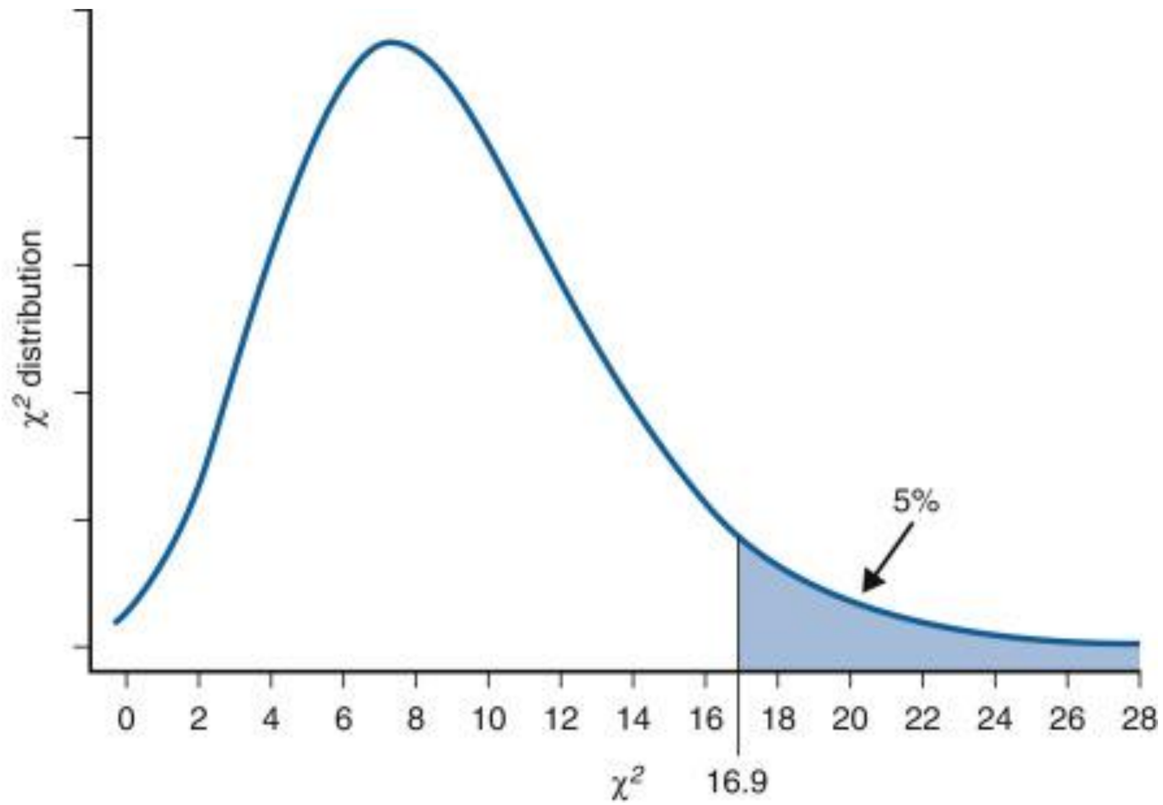
The sample variance $(\boldsymbol{s^2})$ follows a chi-squared distribution with $(N-1)$ degrees of freedom, which allows us to build confidence intervals for the true variance $(\boldsymbol{\sigma^2})$.

$$(N-1)\frac{\boldsymbol{s^2}}{\boldsymbol{\sigma^2}} \sim \chi^2_{N-1}$$

Chi-Squared distribution – Definition (χ^2)

Properties:

- $E(\chi^2) = n$
- $\sigma^2(\chi^2) = 2n$
- $\frac{\chi^2 - n}{\sqrt{2n}} \xrightarrow{n\to\infty} N(0,1), when\ n > 50$



Chi-Squared distribution – Confidence Interval for σ^2

In the **chi-squared distribution tables**, it is possible to find two values g_1 and g_2 such that:

$$P(g_1 < \chi^2_{N-1} < g_2) = 1 - \alpha$$

$$\downarrow$$

$$P\left((N-1)\frac{s^2}{g_2} < \chi^2_{N-1} < (N-1)\frac{s^2}{g_1}\right) = 1 - \alpha$$

$$\downarrow$$

$$IC_{\sigma^2} = \left[(N-1)\frac{s^2}{g_2}, (N-1)\frac{s^2}{g_1}\right] \text{ and } IC_{\sigma} = \left[\sqrt{(N-1)\frac{s^2}{g_2}}, \sqrt{(N-1)\frac{s^2}{g_1}}\right]$$

Proportions - Confidence Interval for p

We introduce a new parameter, the sample proportion ($\hat{\boldsymbol{p}}$):

$$\hat{p} = \frac{x}{n}$$

Where x is the number of successes and n the sample size. For large samples, it's typically based on the **normal approximation**:

$$IC_{\hat{p}} = [\hat{p} + z_{\frac{\alpha}{2}} \cdot \frac{\sqrt{\hat{p} * (1 - \hat{p})}}{\sqrt{n}}, \hat{p} - z_{\frac{\alpha}{2}} \cdot \frac{\sqrt{\hat{p} * (1 - \hat{p})}}{\sqrt{n}}]$$

Proportions – Calculation of the sample size (n)

From the $IC_{\hat{p}}$ formula the width of the interval depends on:

$$margin\ error(E) = z_{\frac{\alpha}{2}} \cdot \frac{\sqrt{p*(1-p)}}{\sqrt{n}}$$

Where p is the population proportion, sometimes not known.
When solving n:

$$n = \frac{z_{\frac{\alpha}{2}} \cdot (p*(1-p))}{E^2}$$

$p*(1-p)$ **is maximized at 0.5**, giving the largest required sample size

Case Study

A week ago, CITRUS S.A. purchased an orange bagging machine. When it was purchased, it was calibrated to produce bags weighing an **average of 2000 g.**

At present, the company is unsure whether the machine is still producing bags of that average weight. If it were certain that the machine is not properly calibrated, it would need to recalibrate it, which would require halting production, with the associated costs.

Case Study

To check that they are working properly, a random sample of 15 packets is taken, yielding the following weights:

1989, 2015, 1962, 2013, 1983, 1989, 1992, 2011,

1958, 2023, 1980, 1977, 1994, 2017, 2001

Descriptive statistics

$$\bar{x} = 1993.6$$
$$s = 19.8$$

Is there any evidence that $\bar{x}$ differs significantly from 2000, and should the filling machine therefore be readjusted?

Within what range can it be stated with reasonable certainty that the value of $\bar{x}$ lies? And s?

Case Study

Hypothesis testing with $\alpha = 0.05$

$$\begin{cases} H_0: \bar{x} \approx 2000 \rightarrow Not\ recalibrating\ the\ machine \\ H_1: \bar{x} \neq 2000 \rightarrow Recalibrating\ the\ machine. \end{cases}$$

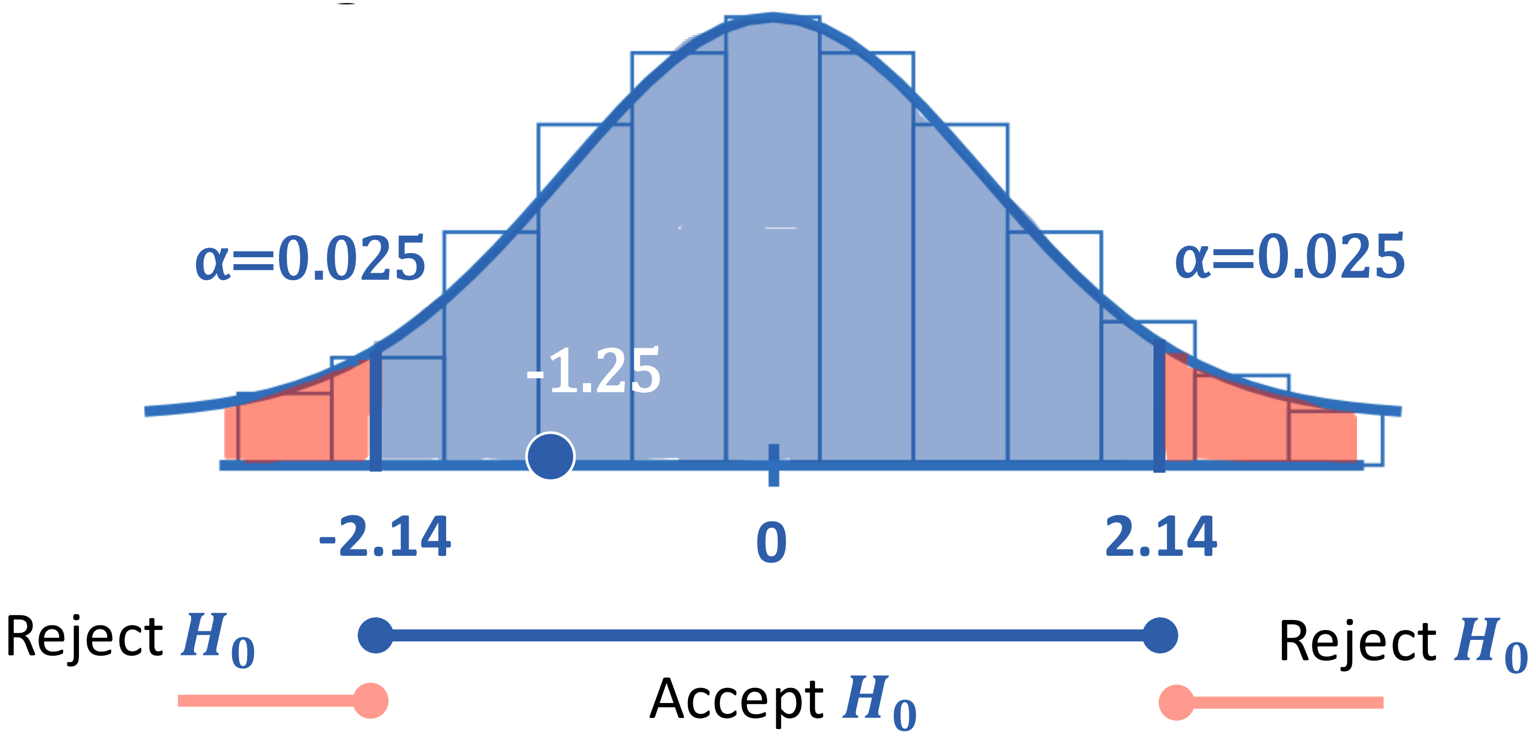
t- Student hypothesis testing

$$P(t_{15-1} > 2.14) = 0.05$$

We calculate the t_{15-1} statistic:

$$\left|\frac{\bar{\boldsymbol{x}} - \mathbf{2000}}{\boldsymbol{s}/\sqrt{\boldsymbol{N}}}\right| = \left|\frac{1993.6 - \mathbf{2000}}{19.8/\sqrt{\mathbf{15}}}\right| = -\mathbf{1.25}$$

Case Study



P-value calculation

$$P(t_{15-1} < -1.25) * 2 \approx 0.1 * 2 = 0.2$$

$$\boldsymbol{p - value = 0.2 > 0.05}$$

We can accept H_0: $\bar{x} \approx 2000$ because:

- $\boldsymbol{-1.25 \in [-2.14, 2.14]}$
- $\boldsymbol{0.2 > 0.05}$

Confidence Interval for m

$$1993.6 \pm 2.14 \cdot \frac{19.8}{\sqrt{15}} \to IC_m = [1982.7, 2004.5]$$

Population means ranging from 1982.7 to 2004.5 are consistent with our sample at a 95% confidence level. Outside this range, the mean is so far removed that it would not be consistent with our data, as the difference cannot be attributed to sampling error.

We can accept $H_0: \bar{x} \approx 2000$ because:

$$2000 \in [1982.7, 2004.5]$$

Confidence Interval for $\boldsymbol{\sigma^2}$

$$IC_{\sigma^2} = \left[(N-1)\frac{s^2}{g_2}, (N-1)\frac{s^2}{g_1}\right] \rightarrow IC_{\sigma^2} = \left[(\mathbf{15}-\mathbf{1}) * \frac{\mathbf{19.8^2}}{\mathbf{26.1}}, (\mathbf{15}-\mathbf{1}) * \frac{\mathbf{19.8^2}}{\mathbf{5.63}}\right]$$

The confidence interval would finally be:

$$IC_{\sigma^2} = [210.28\,, 974{,}87]$$

$$IC_{\sigma} = [14.5\,, 31.2]$$

Case Study

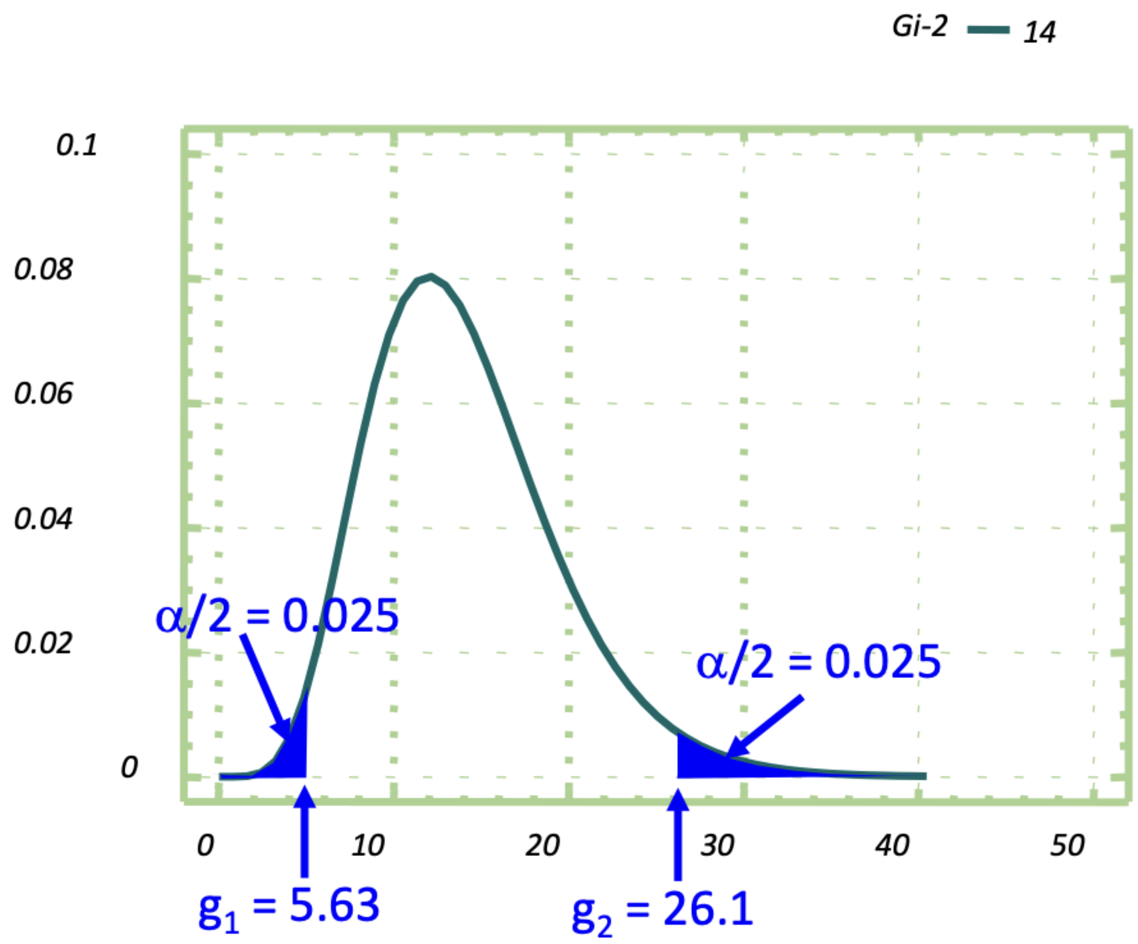


TABLA CHI-CUADRADO $P(\chi^2_n \leq x) = \alpha$

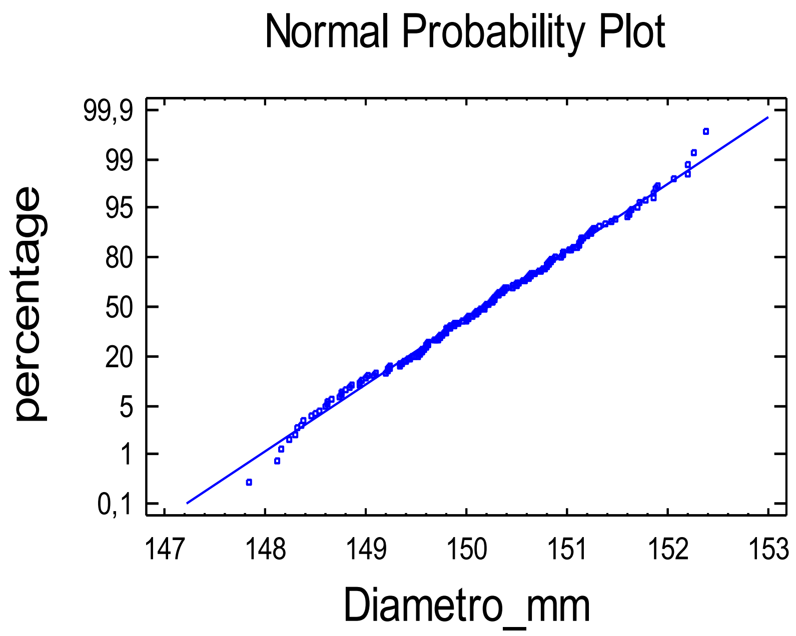
n\α	0,0010	0,0025	0,0050	0,0075	0,0100	0,0250	0,0500	0,0750	0,1000	0,2000	0,4000
1	0,0000	0,0000	0,0000	0,0001	0,0002	0,0010	0,0039	0,0089	0,0158	0,0642	0,2750
2	0,0020	0,0050	0,0100	0,0151	0,0201	0,0506	0,1026	0,1559	0,2107	0,4463	1,0217
3	0,0243	0,0449	0,0717	0,0944	0,1148	0,2158	0,3518	0,4720	0,5844	1,0052	1,8692
4	0,0908	0,1449	0,2070	0,2555	0,2971	0,4844	0,7107	0,8969	1,0636	1,6488	2,7528
5	0,2102	0,3075	0,4117	0,4896	0,5543	0,8312	1,1455	1,3937	1,6103	2,3425	3,6555
6	0,3811	0,5266	0,6757	0,7838	0,8721	1,2373	1,6354	1,9415	2,2041	3,0701	4,5702
7	0,5985	0,7945	0,9893	1,1276	1,2390	1,6899	2,1673	2,5277	2,8331	3,8223	5,4932
8	0,8571	1,1043	1,3444	1,5124	1,6465	2,1797	2,7326	3,1440	3,4895	4,5936	6,4226
9	1,1519	1,4501	1,7349	1,9318	2,0879	2,7004	3,3251	3,7847	4,1682	5,3801	7,3570
10	1,4787	1,8274	2,1559	2,3809	2,5582	3,2470	3,9403	4,4459	4,8652	6,1791	8,2955
11	1,8339	2,2321	2,6032	2,8556	3,0535	3,8157	4,5748	5,1243	5,5778	6,9887	9,2373
12	2,2142	2,6612	3,0738	3,3527	3,5706	4,4038	5,2260	5,8175	6,3038	7,8073	10,1820
13	2,6172	3,1119	3,5650	3,8697	4,1069	5,0088	5,8919	6,5238	7,0415	8,6339	11,1291
14	3,0407	3,5820	4,0747	4,4044	4,6604	5,6287	6,5706	7,2415	7,7895	9,4673	12,0785
15	3,4827	4,0697	4,6009	4,9551	5,2293	6,2621	7,2609	7,9695	8,5468	10,3070	13,0297
16	3,9416	4,5734	5,1422	5,5202	5,8122	6,9077	7,9616	8,7067	9,3122	11,1521	13,9827
17	4,4161	5,0917	5,6972	6,0984	6,4078	7,5642	8,6718	9,4522	10,0852	12,0023	14,9373
18	4,9048	5,6233	6,2648	6,6886	7,0149	8,2307	9,3905	10,2053	10,8649	12,8570	15,8932

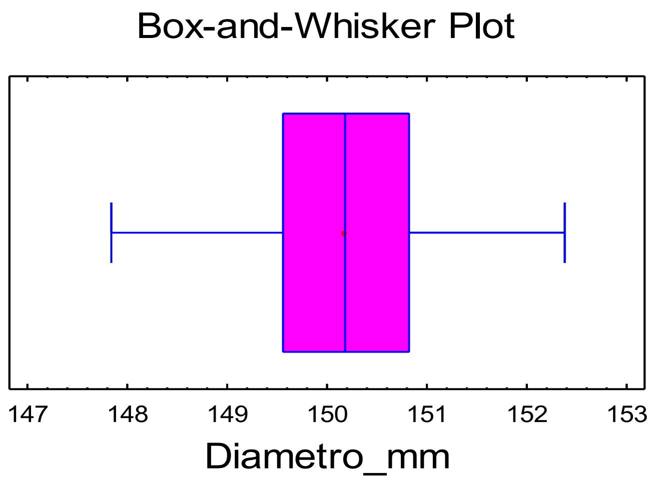
Exercise 1

A company's quality department is analysing a batch of materials from a supplier following the recent detection of certain problems. The variable under investigation is the diameter of the parts.

Exercise 1

The following tables and graphs have been produced from the analysis of a sample of 100 items:





Summary Statistics for Diametro_mm

Count = 100
Average = 150,16
Median = 150,183
Variance = 0,890275
Standard deviation = 0,943544
Minimum = 147,848
Maximum = 152,386
Range = 4,538
Lower quartile = 149,562
Upper quartile = 150,819
Interquartile range = 1,257
Skewness = -0,0441464
Stnd. skewness = -0,254879
Kurtosis = -0,360797
Stnd. kurtosis = -1,04153
Coeff. of variation = 0,628358%

Exercise 1

a) Can the diameters of the parts be assumed to follow a normal distribution? Explain your answer.

b) Based on the sample observed, can the supplier's claim that the average diameter of their parts is 150 mm be accepted, given a Type I risk of 5%? (State the hypotheses to be tested and the acceptance and rejection regions for the test)

Exercise 2

The net weight per sachet of a brand of powdered soup is normally distributed, with a mean of 565 g and a standard deviation of 15 g.

a) If a sample of nine sachets is selected at random from the production run, what is the probability that the sample mean will be between 555 and 575 g? What does this result indicate in practice?

b) How many sachets of soup would need to be sampled for the probability in the previous section to be 0.9906?

c) A change has been made to the production process and the aim is to investigate whether this has affected the mean net weight. To this end, a sample of 5 sachets has been taken from the production run following the process change. The values are shown below: **580 550 555 570 546**

Can it be stated with a Type I risk of 10% that the change in the process has affected the mean net weight?

Exercise 3

During a quality control process carried out at a particular mobile phone company, 61 mobile phones were selected at random, and it was found that 6 of them had manufacturing defects. Determine a 95% confidence interval for the proportion of defective mobile phones in the manufacturing process.

Exercise 4

A survey is to be conducted to estimate the percentage of customers satisfied with the service received from a company's Customer Service department. What is the minimum sample size N required if the probability of making an error of more than 5% points when estimating a given proportion p is to be less than 1%?

Exercise 5

An average radon concentration in the air of 370 Bq/m³ has been detected at a wastewater pre-treatment plant. As this concentration exceeds the reference level for workplaces, regular measurements are carried out to limit workers' exposure.

In a given month, 10 measurements are taken, yielding the following data:

340.5	356.8	520.6	654.1	390.2
296.3	248.9	307.5	342.1	465.8
$\bar{x} = 402.28$				
s=112.63				

Exercise 5

a) Is there statistical evidence to conclude that the average radon concentration increased during that month? Should we set the Type I risk at 0.01?

b) What would be the approximate value of the Type II risk if the average concentration were actually 410 Bq/m^3?

c) Construct a 95% confidence interval for the population standard deviation.